The NGT GPU allocation problem, measured

30 days of cluster telemetry, and what a policy would change

Felice Pantaleo (CERN)
July 2026

Method

Data

Pod lifecycle, requests, placement, cordons and per-GPU utilization from cluster monitoring, 30 days at 5 min resolution.

Scope

1571 GPU/MIG requests from 114 users on the on-premise pools. STEAM Academy accounts and cloud T4 nodes excluded.

Attribution

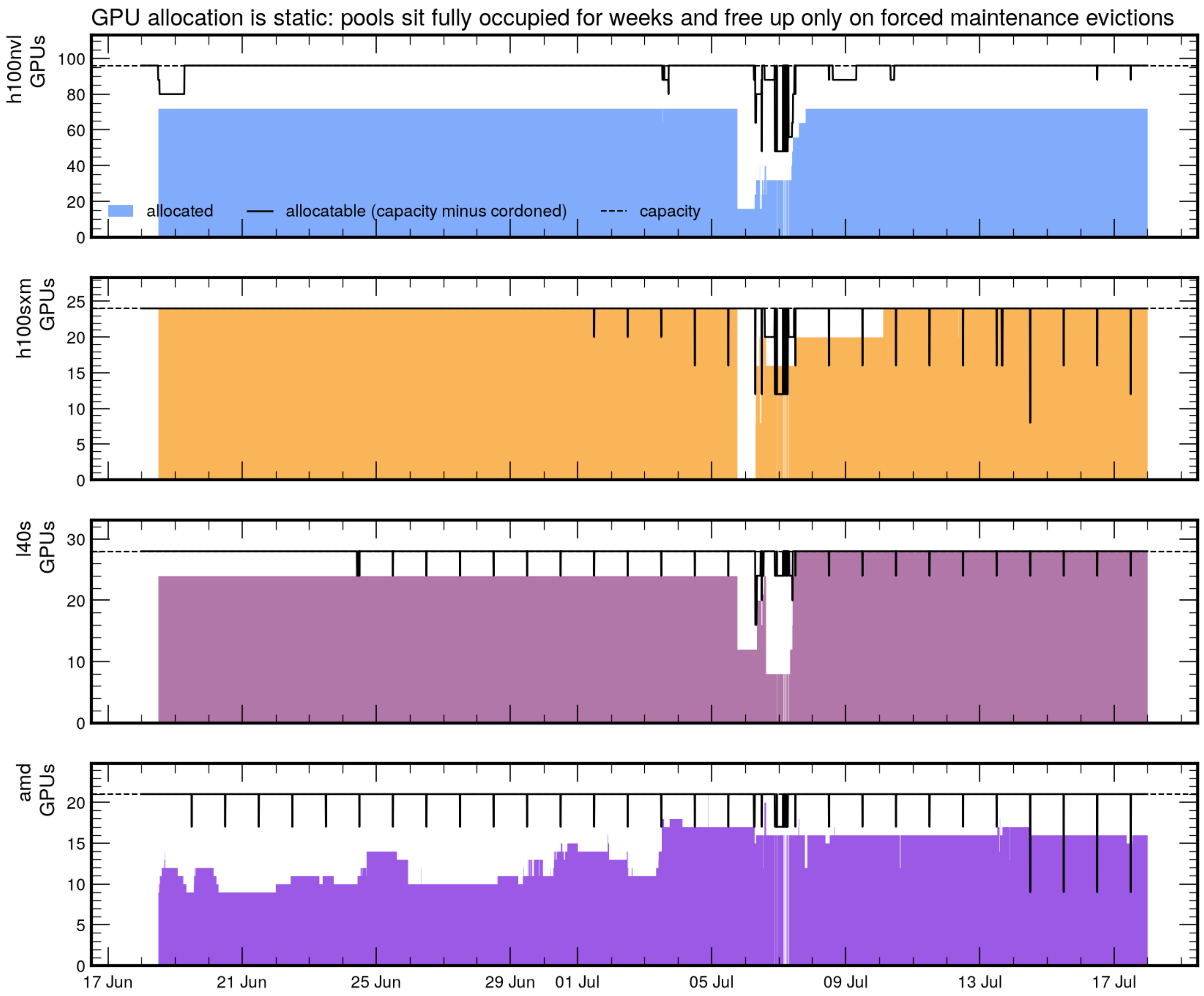
Users mapped to WP1 to WP4 via the project roster, department rules and
manual classification: 99.4% of GPU-hours attributed.

A request = one pod asking for GPUs. Wait = creation to scheduling. Idle = GPU utilization below 5%.

NGT resource allocation

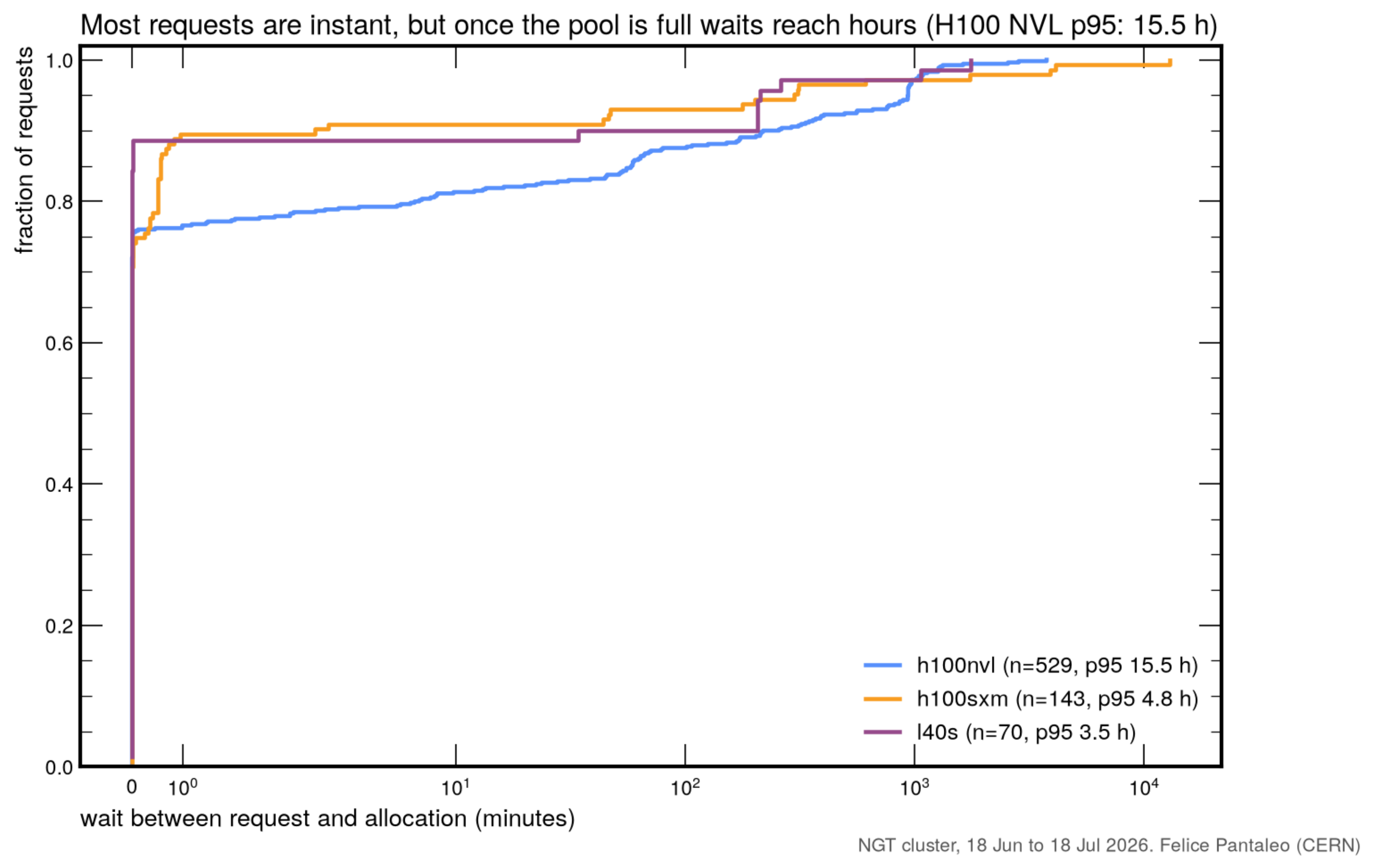
Static allocation

- All pools at their effective ceiling around the clock
- No diurnal variation: allocation follows ownership, not workload
- Only the forced maintenance evictions of 6 to 8 July freed capacity

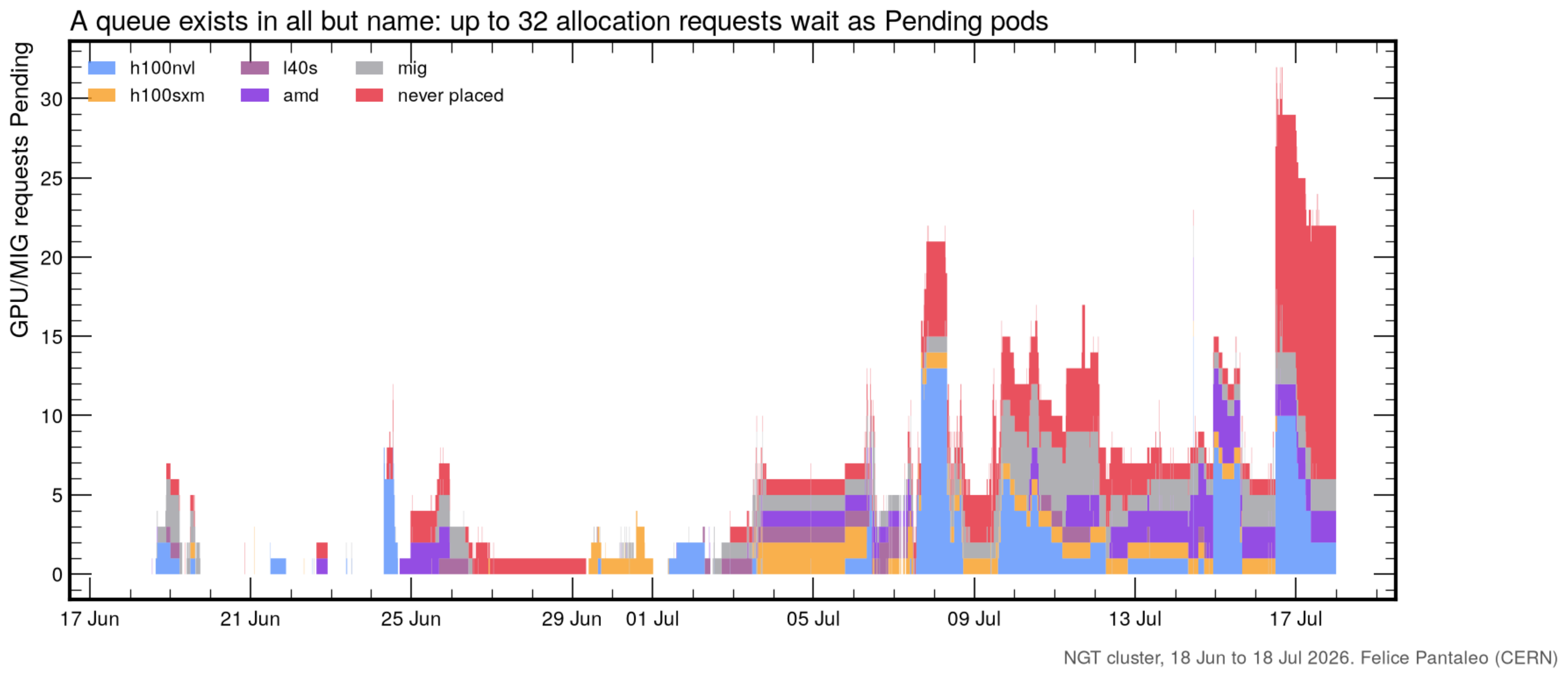


NGT cluster, 18 Jun to 18 Jul 2026. Felice Pantaleo (CERN)
NVL: MIG-enabled GPUs are not visible to DCGM, so the full-GPU ceiling sits below the 96-GPU capacity line. AMD from user-pod accounting (no DCGM).

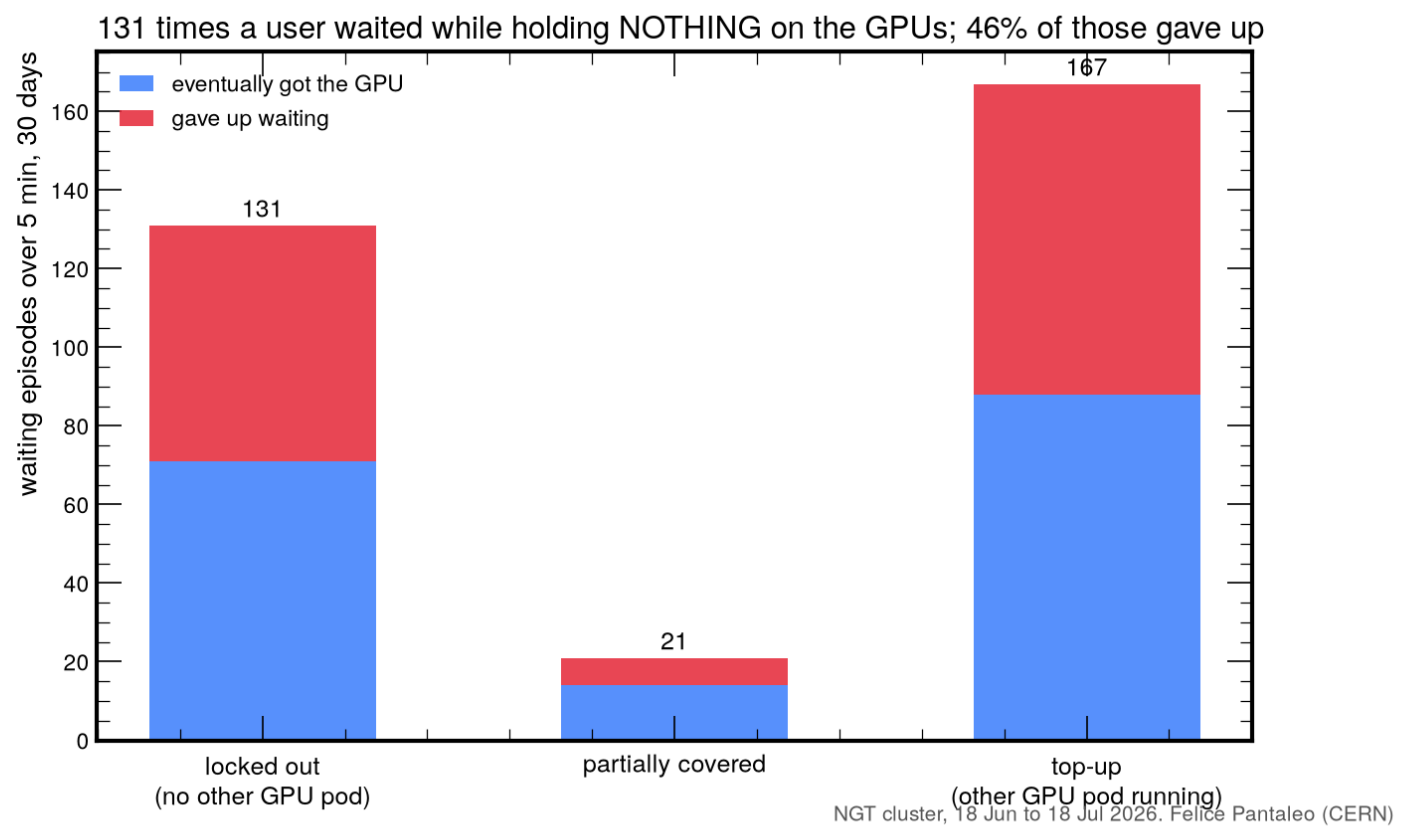
Waiting times



The Pending queue



Lockouts

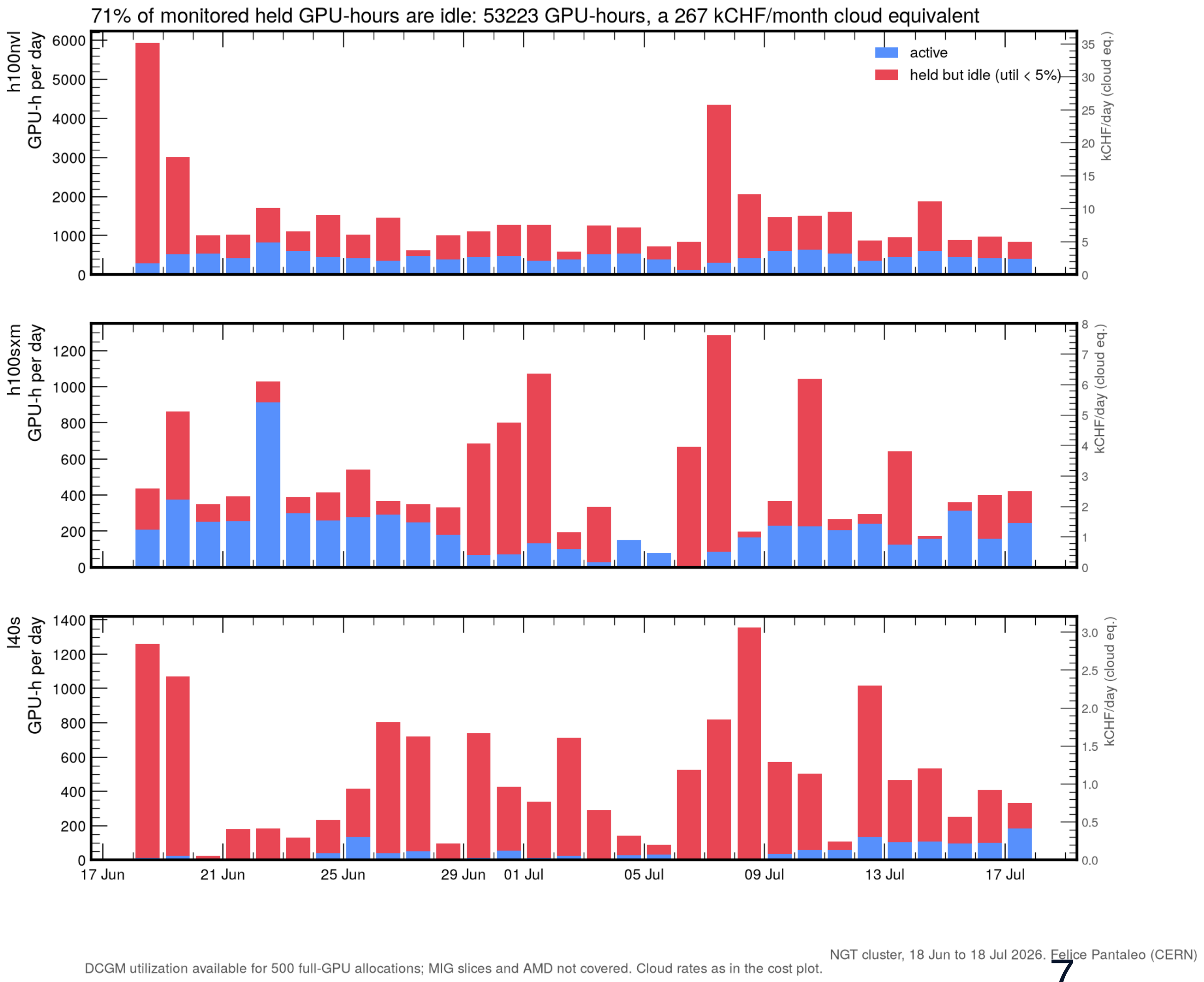


Over half of all queue pressure is top-up demand from users already running.

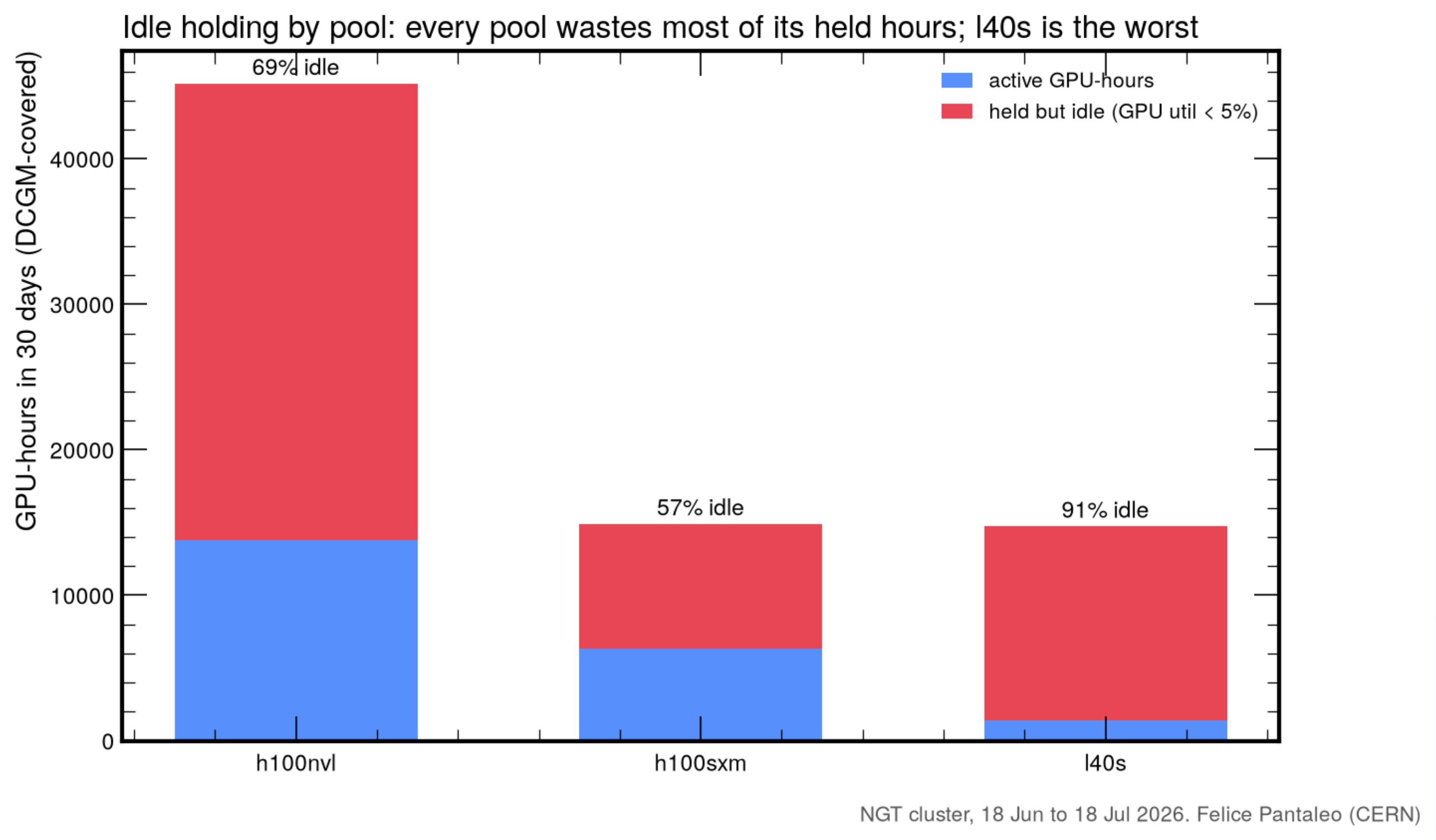
NGT resource allocation

Idle held GPU-hours

- 53 000 GPU-hours parked in 30 days
- 267 kCHF/month cloud equivalent
- Right axes: kCHF/day per pool
- Continuous idle baseline on NVL and L40S; bursty parked episodes on SXM



Idle share by pool

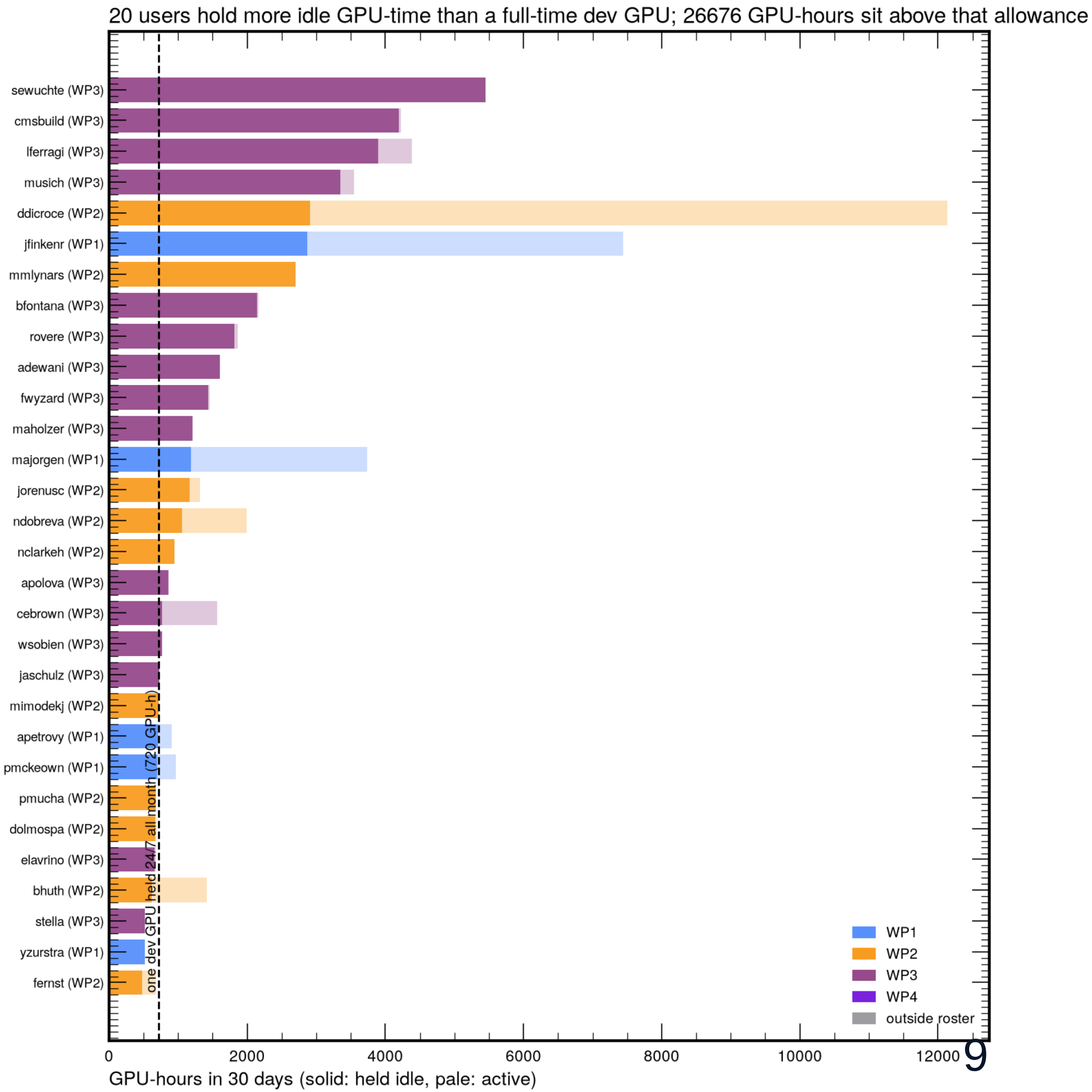


The L40S pool is idle for 91% of its DCGM-covered held hours.

NGT resource allocation

Heavy idle holders

- Solid = held idle, pale = active
- Dashed line: one dev GPU held 24/7 all month (720 GPU-h), the legitimate
  P1 allowance
- **20 users** exceed it
- **26 700 GPU-hours** sit above it
- The consumption and idleness rankings select different users: the top
  consumer is almost entirely active

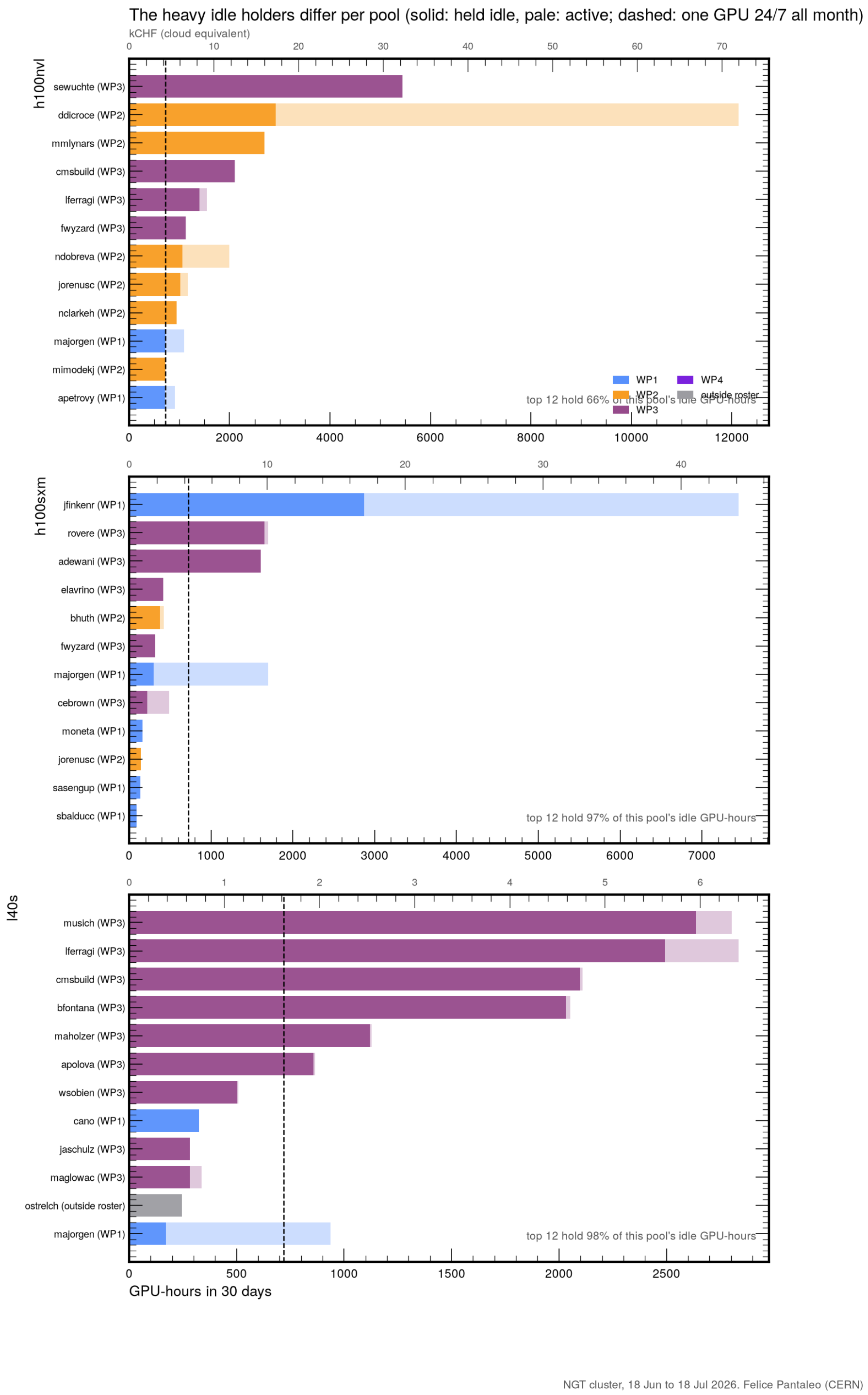


NGT cluster, 18 Jun to 18 Jul 2026. Felice Pantaleo (CERN)

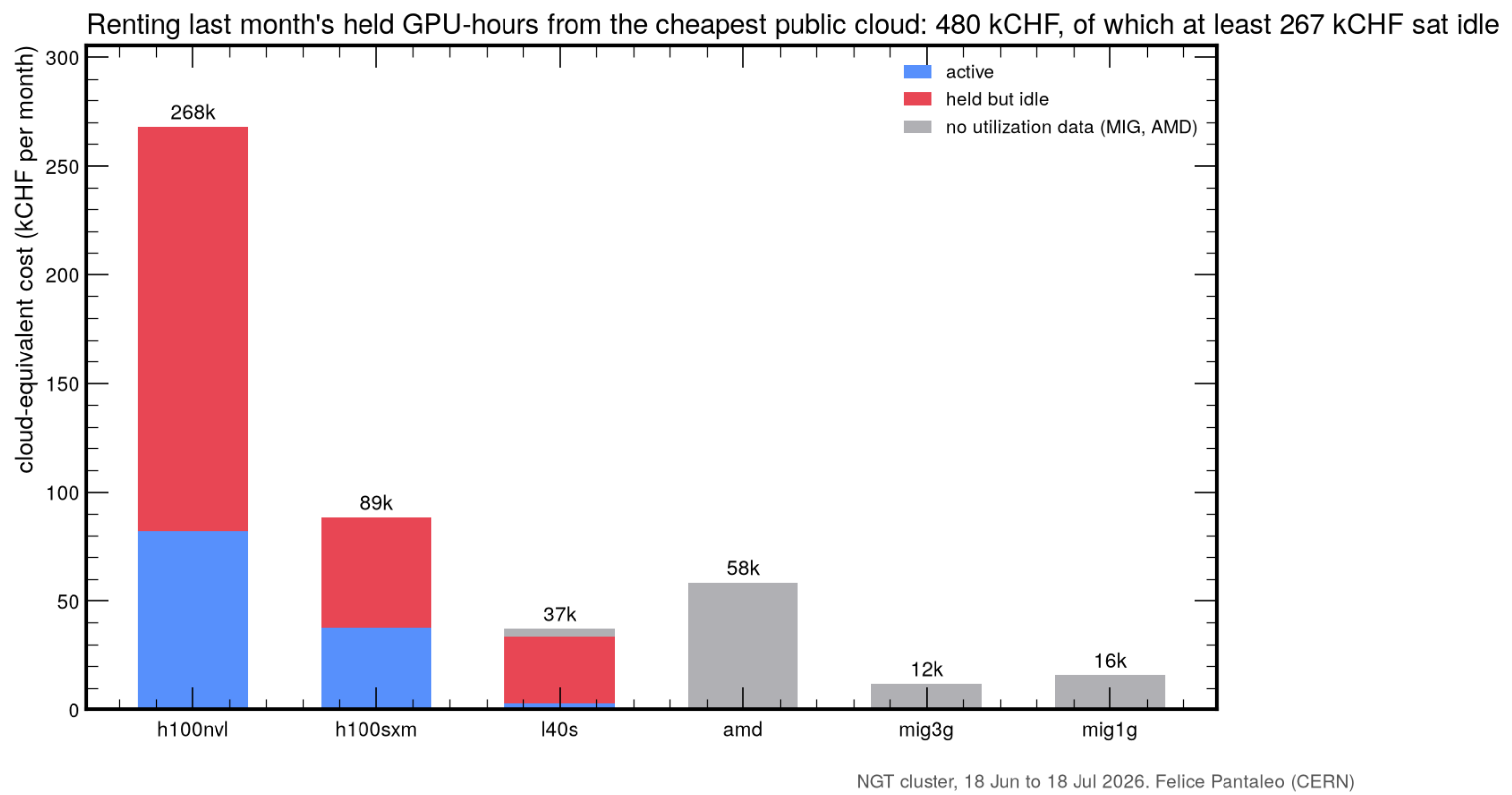
NGT resource allocation

Idle holders by pool

- NVL: broad behavior, top 12 hold 66% of idle
- SXM: three users own the idle hours
- L40S: 12 users own 98% of the idle hours
- The one-session rule and batch-only multi-GPU bound all of this by construction



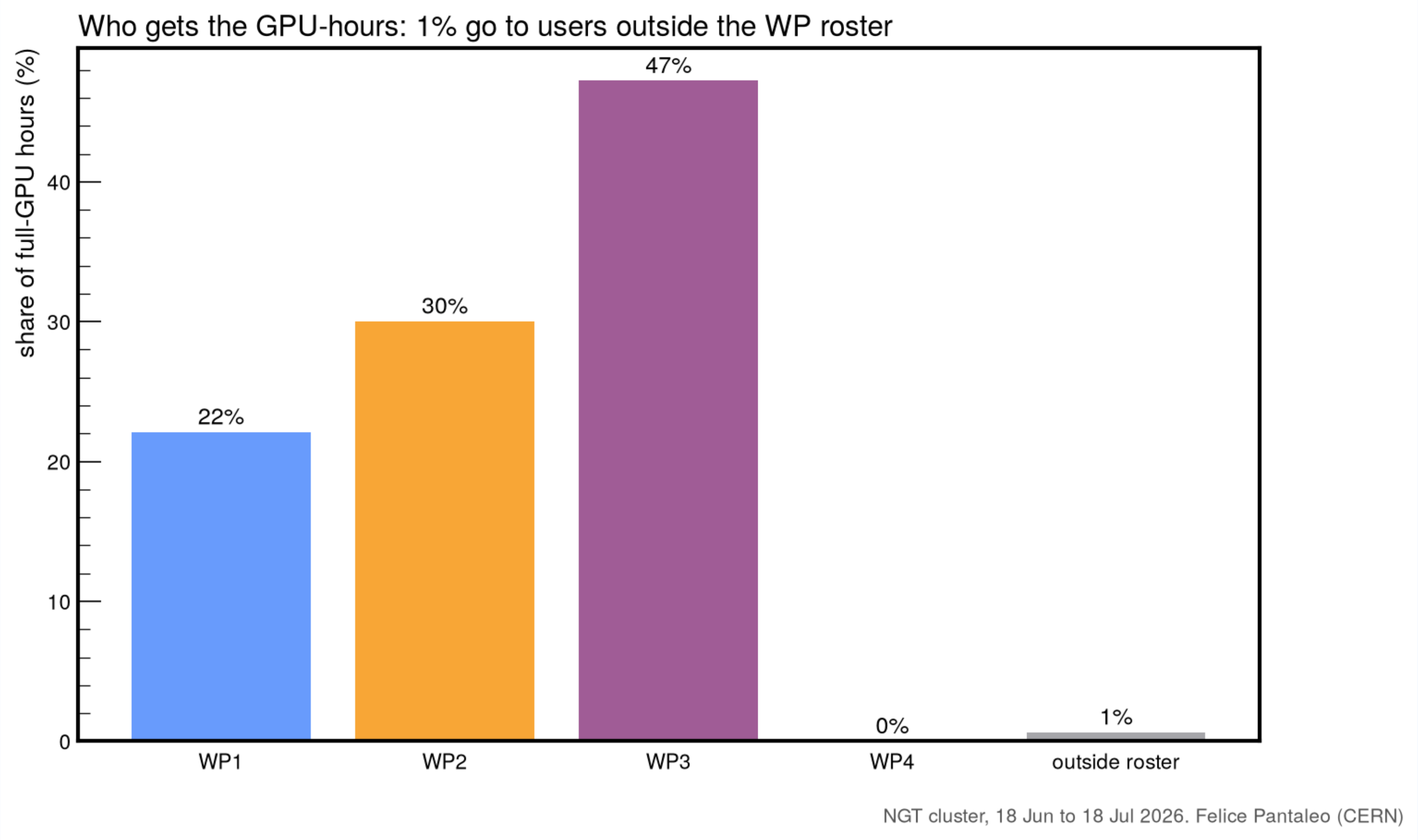
Cloud-equivalent cost



On-demand, cheapest of AWS/GCP, July 2026: H100 6.88 USD/GPU-h (AWS p5.48xlarge; GCP A3 10.98), L40S 2.62 USD/GPU-h (AWS g6e.12xlarge); MIG priced as H100 fractions (1/2, 1/7); MI300X not offered by either, priced H100-class. CHF at 0.862 (NGT budget rate). Egress, storage and CPU-only nodes excluded.

Rental equivalent: 480 kCHF/month, of which at least 267 kCHF idle

GPU-hours by WP



WP2 is on its 30% target organically; WP3 runs at 47% (CMS production

The real month, replayed

Same 30 days, same requests, replayed through the proposed policy:

	today (FCFS)	proposed
wait p95	557 min	**0 min**
requests satisfied	96.7%	**99.4%**
dev sessions within 15 min	91%	**98%**
NVL idle-held GPU-hours	35 800	**13 600**
running work terminated by the system	none	**none**

> **Policy: one interactive session per member (a new session supersedes the previous one); multi-GPU beyond a 96 GPU-h/month interactive allowance runs as batch behind a WP fair-share priority queue. Fixed-behavior replay, identical requests for every policy; the FCFS baseline reproduces observed waits (median exact, p95 within factor 2).**

Proposal to the PMC

One interactive session

One single-GPU session per member, guaranteed and always available; opening a new one replaces the old.

Multi-GPU is batch

Submitted, executed, exits at completion, behind a priority queue with
WP fair share. A 96 GPU-h/month interactive allowance covers debugging.

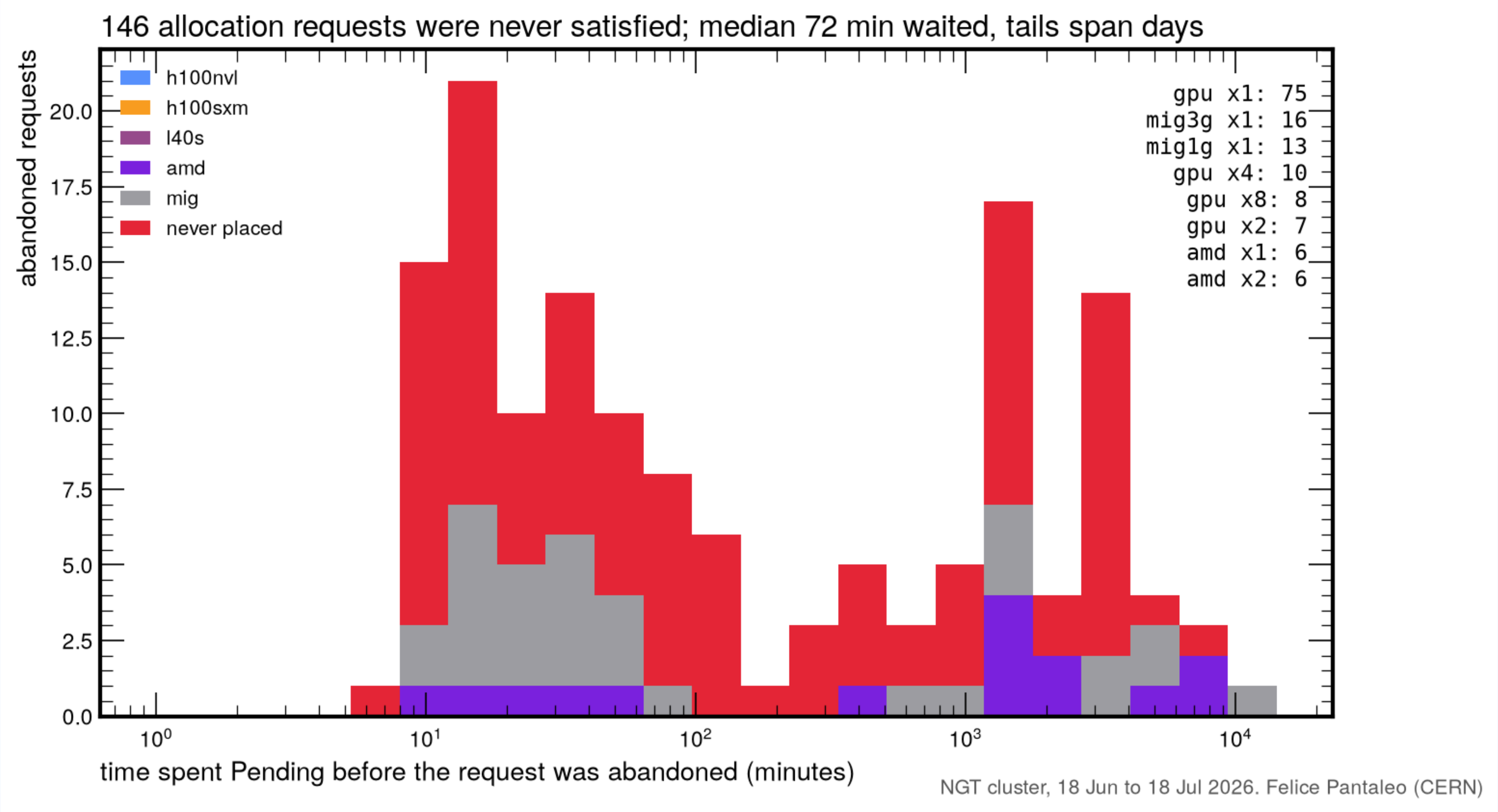
Account and adjust

GPU time charged to WPs with model factors. All parameters simulated and tunable on the real trace.

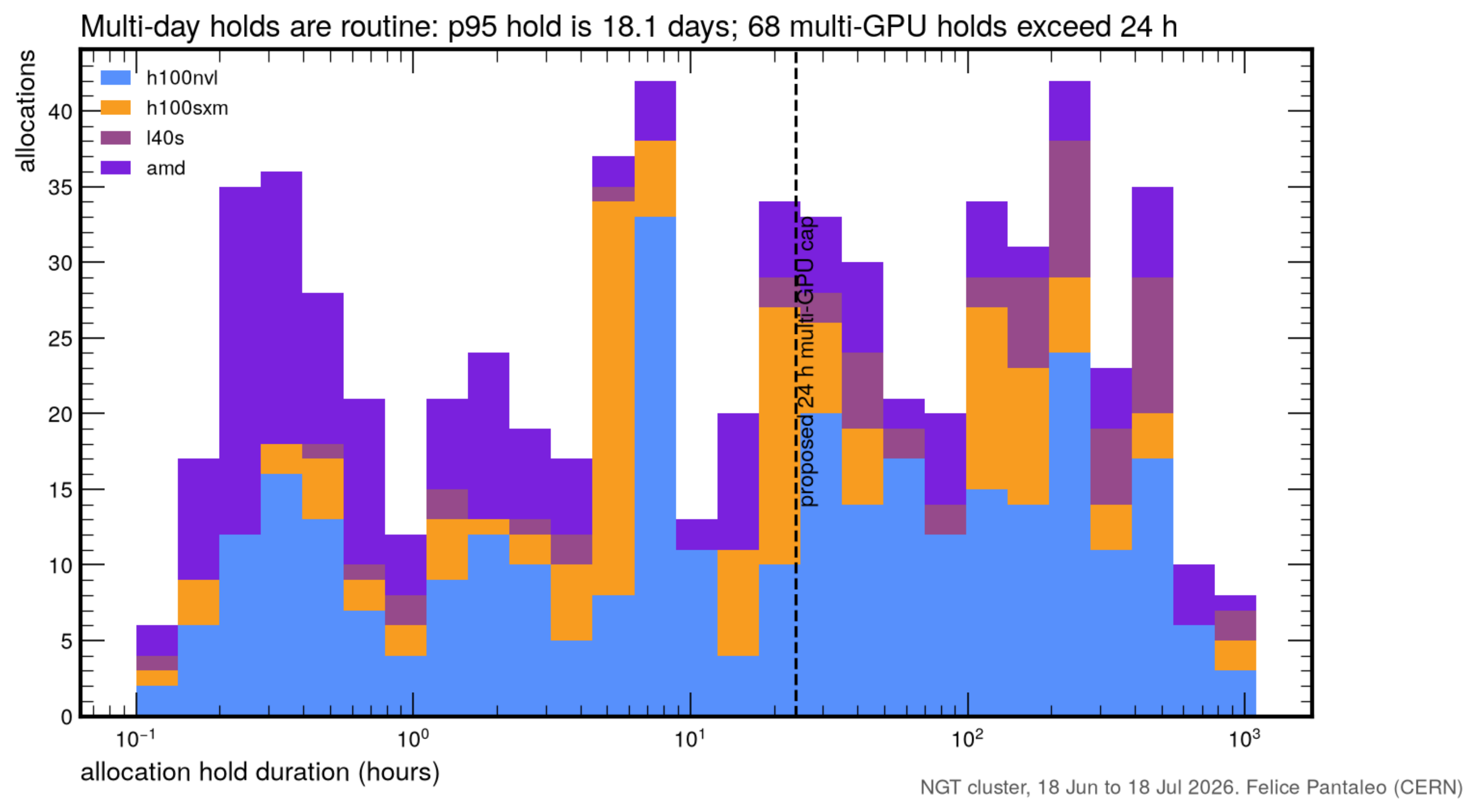
Implementable with Kueue plus one admission webhook and a small accounting controller. The scheduler terminates no running work.

Backup

Backup: unsatisfied requests

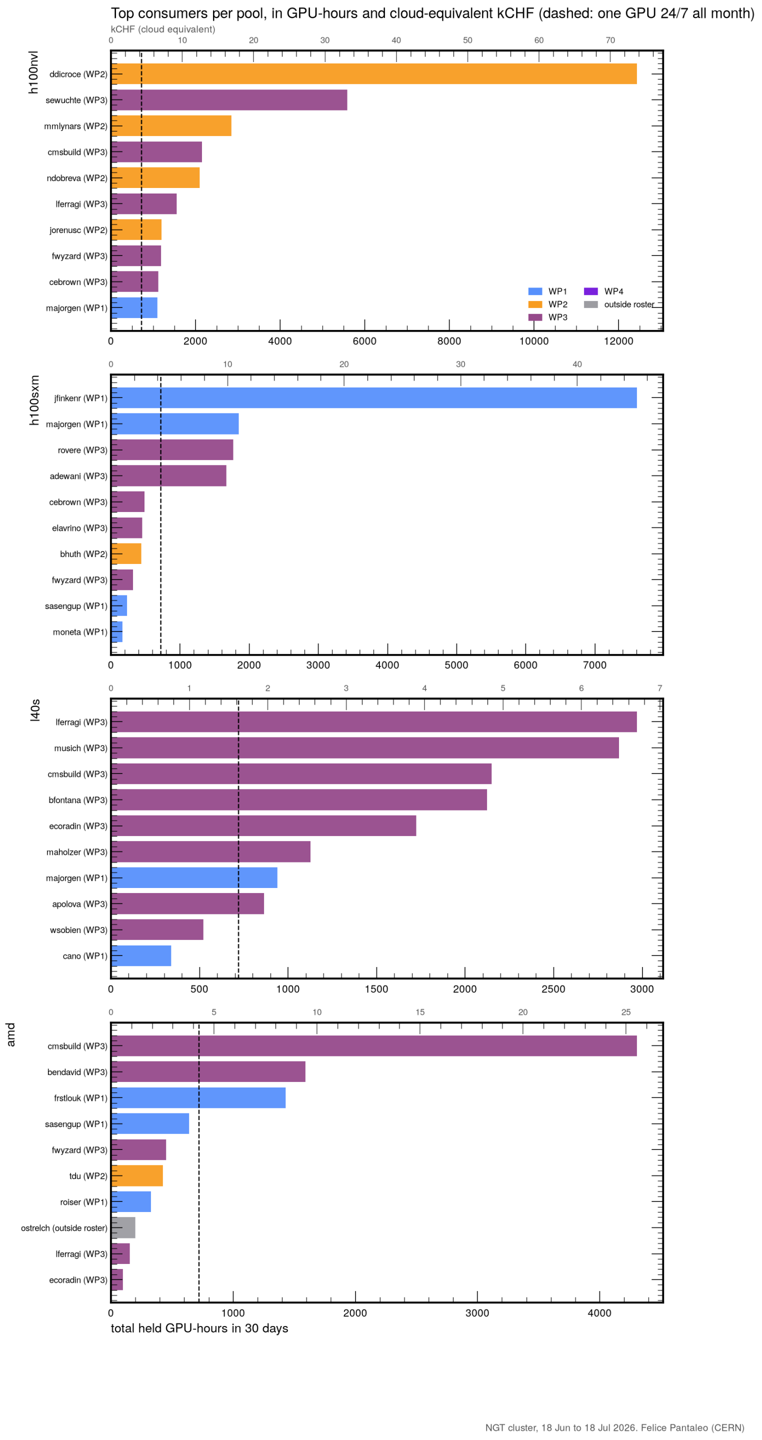


Backup: hold durations



Backup: top consumers per pool

- Secondary axis: cloud-equivalent kCHF, cheapest of AWS/GCP, July 2026
- Dashed: one GPU 24/7 all month
- 31 users held more than one GPU-month



Backup: pools and cordons